

5-Mark Questions & Answers — MPCA Unit 1: Overview of Computer Architecture & Organization

Q1. Explain the Von Neumann architecture in detail with its advantages and limitations.

1. Detailed Concept

The **Von Neumann Architecture** is based on three fundamental principles:

- **Stored-Program Concept:** Instructions and data share the same unified primary memory space.
- **Sequential Execution:** Instructions are fetched and executed one after another, controlled by a Program Counter (**PC**).
- **Functional Components:** Central Processing Unit (ALU + CU), Main Memory, Input/Output interface.

2. Key Advantages

- **Flexibility:** Program code can be dynamically loaded, modified, and re-executed without altering hardware wiring.
- **Simplified Hardware Design:** Uses a single memory controller and single system bus structure.

3. Limitations (Von Neumann Bottleneck)

- **Shared Bus Contention:** The CPU cannot fetch an instruction and read/write data simultaneously over the same bus system.
- **Speed Mismatch:** CPU processing speed scales much faster than RAM access latency, creating CPU stall cycles (idle wait states).

Q2. Explain all addressing modes of the 8086 microprocessor with examples.

1. **Immediate Mode:** Operand is a constant value inside instruction.
 - `MOV CX, 1234H`
2. **Register Direct Mode:** Operand resides in CPU register.
 - `MOV AX, BX`
3. **Direct Memory Mode:** Operand memory offset specified directly.
 - `MOV AX, [1000H]`
4. **Register Indirect Mode:** Address stored inside pointer register (**BX** , **BP** , **SI** , **DI**).
 - `MOV AX, [BX]`
5. **Based Addressing Mode:** Effective Address = Base Register (**BX** or **BP**) + Displacement.
 - `MOV AX, [BX + 08H]`
6. **Indexed Addressing Mode:** Effective Address = Index Register (**SI** or **DI**) + Displacement.
 - `MOV AX, [SI + 10H]`
7. **Based-Indexed Mode:** Effective Address = Base Register + Index Register.
 - `MOV AX, [BX + SI]`

8. **Based-Indexed with Displacement Mode:** Effective Address = Base Register + Index Register + Displacement.

- `MOV AX, [BX + SI + 04H]`

Q3. Discuss the performance measures of computer architecture (CPU time, clock cycles, MIPS, benchmarks) in detail.

1. CPU Execution Time Formula

```
CPU Time = Instruction Count (IC) × CPI × Clock Cycle Time (T)
CPU Time = (Instruction Count × CPI) / Clock Rate (f)
```

2. Clock Cycles & CPI

- **Clock Rate (f):** CPU clock frequency measured in GHz.
- **CPI (Cycles Per Instruction):** Average number of clock cycles required to execute one instruction.

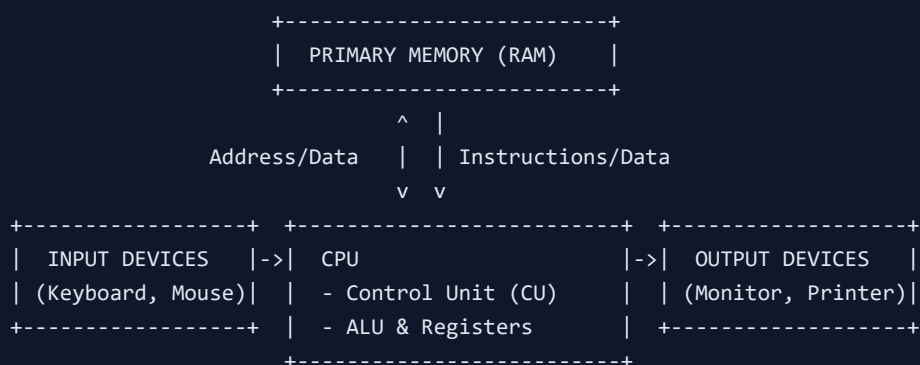
3. MIPS (Million Instructions Per Second)

```
MIPS = Instruction Count / (Execution Time × 10^6)
MIPS = Clock Rate / (CPI × 10^6)
```

4. Benchmarks

Standardized suite of test programs (e.g., **SPEC CPU2017**, **Geekbench**, **Linpack**) used to evaluate real-world hardware performance across arithmetic, memory bandwidth, and multitasking workloads.

Q4. Explain the basic organization and block-level description of the functional units of a computer.



1. **Input Unit:** Accepts external user data/programs and converts them into binary code.

2. Central Processing Unit (CPU):

- **Control Unit (CU):** Decoding instructions, generating synchronization clock pulses.
- **Arithmetic Logic Unit (ALU):** Performs operations (`+` , `-` , `AND` , `OR` , shift operations).
- **Registers:** High-speed internal storage locations (`PC` , `IR` , `MAR` , `MDR`).

3. **Memory Unit:** Primary RAM storing active programs and working variables.

4. **Output Unit:** Translates processed binary results into human-readable media.

Q5. Compare Harvard architecture with Von Neumann architecture.

Parameter	Von Neumann Architecture	Harvard Architecture
Memory System	Shared single memory for instructions and data.	Separate physical memory blocks for instructions and data.
Bus System	Shared single bus for instruction fetch & data access.	Separate instruction bus and data bus.
Pipelining Efficiency	Lower efficiency due to bus conflict bottlenecks.	Higher efficiency (simultaneous instruction & data access).
Complexity & Cost	Simple structure, lower hardware cost.	Complex memory controller, higher cost.
Applications	General-purpose desktop computers, servers.	DSP processors, Microcontrollers (ARM Cortex, PIC, AVR).